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TÍTULO: Long-Short Term Memory (LSTM) for Context Analysis on Tweets to Provide Insight into Human Trafficking Problem

Trabajo de integración curricular presentado como requisito para la obtención del título de Ingeniero en Tecnologías de la Información

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Dedication

I dedicate this work to all the people who have supported and accompanied me during my career at Yachay Tech. First, I want to thank my father and mother for all the unconditional support they have given me throughout my life, allowing me to complete my studies; without their help, none of this would have been possible. Secondly, I want to thank my girlfriend, Dánely, for always providing emotional support during the highs and lows of writing this work. She always pushed and motivated me to finish my thesis to complete my university career. Thirdly, and no less importantly, I want to thank my best friend, Leo, whom I met when I started my degree, who has become my brother. He has always supported me in all the crazy ideas, projects, outings, and situations during our stay at Yachay Tech. He is an exemplary person and a life reference for me; thank you for everything. Fourthly, to all my friends whom I discovered over these almost 6 years as Mafer, Jean, Maicol, Cabas, Ariana, and Vincent, who have allowed me to explore the university facet pleasantly and who will always have a place in my heart. Finally, thanks to the professors who have guided me with their knowledge, wisdom, and experience throughout my career, which has allowed me to achieve accomplishments and focus on my goals. To Professor Erick Cuenca, not only for being my thesis tutor but also for being an essential guide for the realization of this and several works that have been published, and especially for the patience he has had with me in carrying out this work, as well as Professor Silvana Escobar, my co-thesis tutor. I am grateful to Professor Israel Pineda, who taught me algorithms and awakened a taste for Computer Science. Professor Eugenio Morocho taught me the basics of artificial intelligence. He allowed me to develop a paper that was published at a conference.

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Mike Bermeo

Resumen

La trata de personas es una problemática que afecta a millones y se ha intensificado con las redes sociales que representa un desafío en la era digital. Redes como X (anteriormente Twitter) y el uso de algoritmos de Aprendizaje Automático e Inteligencia Artificial ofrecen medios para detectar y prevenir estos crímenes. Este trabajo se enfoca en la identificación de tweets relacionados con la trata y el intercambio de material ilícito. Se emplearon modelos de Aprendizaje Automático y Long Short-Term Memory para desarrollar un conjunto de datos específico, extraído de X, que incluye tweets de contenido sexual y otros sospechosos de estar vinculados con la trata. La investigación utilizó métodos variados de representación textual: Bag of Words, Word2Vec + Emoji2Vec y el modelo de embedding ada-002, para capturar la complejidad del lenguaje en estos tweets. Ante el desequilibrio del dataset, se aplicó data augmentation con el modelo gpt3.5-turbo-0125 de OpenAI, enriqueciendo los datos para un entrenamiento más efectivo de los modelos. El modelo basado en Bi-LSTM con embeddings ada-002 y data augmentation demostró ser superior, mostrando eficacia notable en la identificación precisa de tweets relacionados con la trata de personas, lo que representa un avance significativo en la lucha contra esta grave problemática mediante la inteligencia artificial.

Palabras Clave:
Trata de personas, Twitter, Aprendizaje Automático, Long-Short Term Memory, ChatGPT, Clasificación de texto

Abstract

Human Trafficking, a problem affecting millions, has intensified with the advent of social media, representing a challenge in the digital era. Social media like X (before Twitter) and the use of Machine Learning (ML) and Artificial Intelligence algorithms offer means to detect and prevent these crimes. This work focuses on identifying tweets related to Human Trafficking and the exchange of illicit material. ML and Long Short-Term Memory models were employed to develop a specific dataset, extracted from X, which includes tweets with sexual content and others suspected of being linked to Trafficking. The research utilized various textual representation methods, such as Bag of Words, Word2Vec + Emoji2Vec, and the ada-002 embedding model, to capture the complexity of the language in these tweets. Given the imbalance of the dataset, data augmentation was applied using OpenAI's gpt3.5-turbo-0125 model, enriching the data for more effective training of the models. The Bi-LSTM model with ada-002 embeddings and data augmentation proved superior, showing remarkable efficacy in accurately identifying tweets related to Human Trafficking, representing a significant advance in the fight against this serious issue through artificial intelligence.

Keywords:
Human Trafficking, Twitter, Machine Learning, Long-Short Term Memory, ChatGPT, Text Classification

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Chapter 1

Introduction

1.1 Background

Human trafficking is a global problem affecting millions of people worldwide [1]. It is a form of modern slavery that involves the exploitation of individuals for various purposes, including forced labor, sexual exploitation, and organ removal, among others [2]. In recent years, social networks have emerged as a critical platform for human traffickers to operate and reach potential victims [3]. Consequently, there is a growing interest among researchers and policymakers in the role that social networks and other internet-based platforms play as facilitators of this type of crime that affects millions of people globally. Efforts are being made to employ various methodologies to identify and prevent these forms of exploitation.

Technology is being used in increasingly unethical ways. For example, many traffickers use deception techniques to hide their identities and avoid detection.[4]. They can also utilize social networks to trick potential victims, often using emotional manipulation and other tactics to gain their trust [2]. In addition, the vast amount of data generated on social networks can make it challenging to identify and track traffickers and their victims [5].

However, social media can also serve as a valuable tool for raising awareness of human trafficking and gathering support for victims and survivors [6]. Through social media campaigns and the sharing of information and resources, individuals and organizations can take action to combat human trafficking. Additionally, social media provide a space for survivors to share their stories and connect with others, helping to break down isolation and stigma. While social media functionalities have the potential to play a significant role in anti-trafficking efforts, it is essential to recognize its limitations and the need for broader actions to address the root causes of human trafficking and provide support to the victims [7].

In this context, technological advances present new opportunities for the detection and prevention of human trafficking. For instance, data generated on social networks can be utilized to identify patterns and trends in human trafficking activity [3]. Also, Machine Learning (ML) techniques and other data-driven approaches can be employed to create algorithms capable of automatically identifying and flagging suspicious activity on social networks, helping to address this social problem [8]. These technologies can also aid in the

detection of human trafficking advertisements and the identification of relevant keywords used in social media to facilitate these crimes. [9].

1.2 Problem statement

Human trafficking is a critical and pervasive global issue, often hidden in the shadows of society. The rapid proliferation of social media platforms has created an opportunity to identify potential cases of human trafficking and provide timely intervention. This research addresses the critical need for an effective and automated system to detect tweets related to human trafficking, enabling law enforcement agencies, NGOs, and social platforms to take proactive measures.

The primary challenges in this endeavor include the unstructured and noisy nature of social media data, the dynamic language used in tweets, and the subtle and often coded references to human trafficking activities. Existing methods for text classification on social media lack the nuance required for accurate detection. To tackle these challenges, this work proposes utilizing Long Short-Term Memory (LSTM) networks, a type of deep learning model known for its ability to capture sequential patterns and contextual information in textual data.

The key objectives of this research are to develop a robust LSTM-based model for tweet classification, optimize its performance through fine-tuning and hyperparameter optimization, and evaluate its effectiveness in identifying tweets related to human trafficking in real-world scenarios. By addressing these challenges and leveraging LSTM technology, this study aims to significantly improve the accuracy and efficiency of human trafficking detection on social media, thereby contributing to the broader efforts to combat this heinous crime.

1.3 Objectives

1.3.1 General Objective

To develop a deep learning model utilizing Long Short-Term Memory (LSTM) networks for the automated detection and classification of tweets related to human trafficking on X

1.3.2 Specific Objectives

1. *Data Collection and Preprocessing:* Gather a comprehensive dataset of tweets related to human trafficking and preprocess the data to remove noise, irrelevant information, and personally identifiable details.

2. *Dataset:* To construct a comprehensive and annotated dataset for training and evaluating baseline ML models and the LSTM-based tweet classification model by manually collecting and labeling a diverse set of X data, ensuring the inclusion of human trafficking-related tweets and creating a ground truth reference for model assessment.

3. *Emoji Featuring:* To investigate the impact of including emojis as features in the LSTM-based tweet classification model by comparing the model's performance when trained with and without emoji data to determine whether emojis enhance the accuracy and contextual understanding of human trafficking-related tweets.

4. *Text representation:* Explore various techniques for text representation, such as Bag of Words, word embeddings, and document embeddings, to capture nuanced information within tweets.

5. *Training and Validation:* Train the LSTM model on a labeled dataset and validate its performance through techniques like cross-validation and early stopping to prevent overfitting.

1.4 Contribution

This study is one of the few that address human trafficking by analyzing tweets in Spanish. Similar research efforts are lacking, particularly those that employ Long-Short Term Memory (LSTM) models in conjunction with various text representation techniques. Furthermore, a derivative work from this research by Bermeo et al.[10] has been published in the Conference on Information and Communication Technologies of Ecuador (TICEC 2023), focusing on a literature review presented in the state-of-the-art chapter of this work. This study could serve as a foundational framework for analyzing alternative algorithms and frameworks, exploring avenues beyond the ones proposed here, particularly in the context of Spanish language data.

Chapter 2

Theoretical Framework

2.1 Human Trafficking

Human trafficking is characterized by using force or coercion to recruit, transport, and exploit individuals for various purposes, including sexual exploitation, forced labor, and organ removal. It involves the abuse of power or vulnerability and the offering or receiving of payments or benefits in exchange for consent from those in positions of authority. Exploitation can include sexual exploitation, such as prostitution, and other forms of exploitation, including forced labor, slavery, servitude, and organ removal [2].

Social media platforms have become increasingly important in recruiting and grooming potential trafficking victims. For example, traffickers may use social media to lure individuals with false promises of employment or romantic relationships and then exploit them once they have been recruited [2]. In some cases, traffickers may use social media to advertise their victims for sexual exploitation or forced labor or to arrange transportation to different locations [11].

There are several ways in which social media can facilitate human trafficking [2]. First, it allows traffickers to reach a broad audience and target specific demographics, such as young people or those vulnerable due to economic or social circumstances. Second, social media can provide anonymity and secrecy, allowing traffickers to operate without detection. Third, it can be used to obscure the true nature of the exploitation, for example, by presenting it as legitimate work or a consensual relationship.

Combating human trafficking on social media requires a multifaceted approach involving governments, law enforcement agencies, stakeholders, and technology companies. They must work together to fight human trafficking and protect victims' rights. In addition, efforts to raise awareness and educate the public about the risks of human trafficking on social media can help prevent individuals from falling victim to these crimes [2]. Human trafficking is a severe global problem that the proliferation of social media has exacerbated.

2.2 Artificial Intelligence (AI)

Artificial Intelligence (AI) is a branch of computer science and engineering that combines multiple disciplines to develop systems and programs able to execute functions usually needing human cognitive abilities. These functions encompass logical reasoning, finding solutions to problems, acquiring knowledge, interpreting sensory input, comprehending language, and making choices. The concept of AI dates back to ancient history, but it has evolved significantly in recent decades due to advances in computing power, algorithms, and data availability. AI systems have become ubiquitous daily, revolutionizing industries ranging from healthcare and finance to transportation and entertainment.

AI falls into two distinct broad categories: narrow or weak AI and general or strong AI. Narrow AI refers to AI systems that have been developed for a specific task, such as language translation or image recognition. Such systems are excellent at the given task but lack the extensive variety of cognitive capabilities of a human being. By contrast, general AI, frequently found in science fiction, is machines possessing human-level intelligence that can comprehend and execute diverse tasks. Although narrow AI dominates contemporary reality, the creation of general AI is an ongoing long-term objective.

2.3 Machine Learning (ML)

Machine Learning (ML) techniques involve using algorithms and statistical models to analyze and identify patterns of large amounts of data [12]. These techniques have been employed in various fields, including image recognition [13], speech recognition [14], natural language processing [15], and predictive modeling [16]. They can be used as well in the battle against human trafficking. For instance, examining the content of social media posts and finding any that might be about trafficking [17], identifying patterns in the movements of individuals that may indicate trafficking activity [18, 19, 20], detecting anomalies in employment records that could indicate the presence of forced labor [21], predicting the likelihood of individuals becoming victims of trafficking [22, 23], and identifying potential victims before they are exploited [24].

2.3.1 Supervised Learning

Supervised learning is a subcategory of ML that uses labeled training data to predict outputs or classify data into predefined categories. The goal is to build a model to make accurate predictions or classifications based on input data. This is achieved by providing the model with a large set of labeled training examples consisting of input data and the corresponding correct outputs or classifications. The model is then trained to learn the relationship between the input data and the outputs or classifications, using this training data as a guide.

Supervised learning can be used to solve a wide range of problems (see Figure 2.1), including regression, classification, and structured prediction tasks [25]. It is also relatively straightforward to implement, as it requires the availability of labeled training data and a clear objective for the model to optimize [26]. However, one of the limitations of supervised

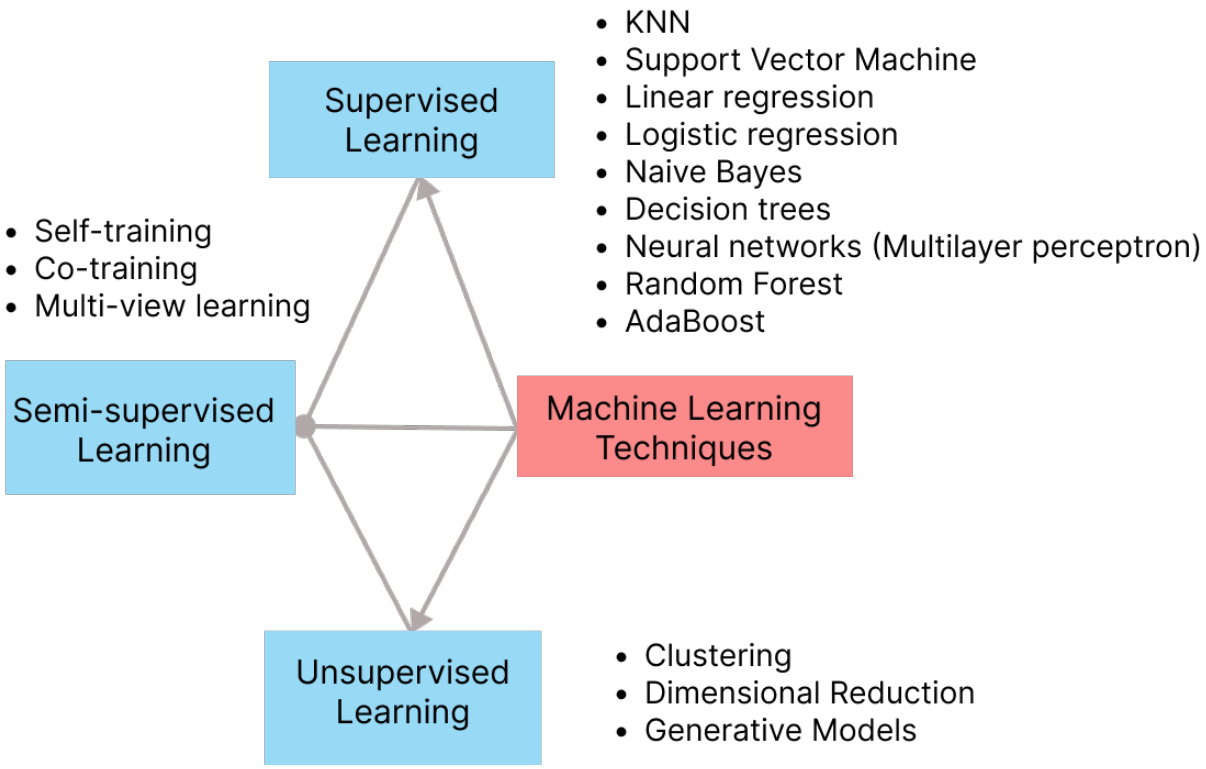

Figure 2.1: Machine learning methods for assessing human trafficking in social media.

learning is that it requires a large amount of labeled data, which can be time-consuming and costly to obtain [27].

To improve the performance of a supervised learning model, it is essential to select the input variables that will be used to train the model and to optimize the model's hyperparameters, which are settings that control the model's behavior. It is also essential to evaluate the model's performance on a separate test dataset to assess its generalization ability and to identify any overfitting or underfitting issues [28, 26].

Considering human trafficking, several types of supervised learning algorithms have been used (see Figure 2.1), including Linear regression, Logistic regression, K-Nearest Neighbors (KNN), Support Vector Machines (SVMs), Naive Bayes (NB), Random Forest (RF), Neural Networks (NN), Decision Tree (DT), AdaBoost, and so on. For example, linear regression [29] predicts a continuous output value based on one or more input variables. Logistic regression [30] is used to classify data into two categories based on input variables. SVMs [17] classify data into multiple categories by finding the hyperplane in a high-dimensional space that maximally separates the different classes.

Supervised learning is a powerful tool widely used to accurately predict outputs or classify data based on labeled training examples. It is a valuable tool for solving many real-world problems, such as Human Trafficking.

2.3.2 Unsupervised Learning

Unsupervised learning is a type of ML in which a model is trained to discover patterns and relationships in a dataset without using labeled training examples [31]. In this learning,

the goal is to find hidden patterns in the data rather than to make specific predictions or classifications [32]. This is achieved by providing the model with a large dataset and allowing it to learn the underlying structure of the data through techniques such as clustering [33] or dimensionality reduction [34].

One of the main advantages of unsupervised learning is that it can extract valuable insights and knowledge from large amounts of unlabeled data. It can also perform data preprocessing and feature extraction tasks, which can be helpful for downstream supervised learning tasks. However, one of the limitations of unsupervised learning is that it can be challenging to interpret the results and to determine the meaning of the discovered patterns and relationships [27].

There are several types of unsupervised learning algorithms (see Figure 2.1), including clustering, dimensionality reduction, and generative models [35]. For instance, Clustering algorithms group data points into clusters based on their similarities. Dimensionality reduction methods reduce the number of dimensions in a dataset by projecting the data onto a lower-dimensional space. Generative models are used to learn the probability distribution of the data and generate new samples similar to the original data [36]. It is also essential to evaluate the model's performance using appropriate evaluation metrics, such as silhouette scores (for clustering) or reconstruction errors (for dimensionality reduction) [37].

To apply k-means clustering, the data is pre-processed to remove irrelevant features and transform them into a format suitable for the algorithm. Next, the number of clusters is chosen using techniques like the elbow method or silhouette analysis. Finally, once the clustering is done, the researcher can examine the resulting clusters to see if they contain any human trafficking activity.

Another unsupervised technique is anomaly detection. This technique works by identifying data points that deviate significantly from the expected behavior of the rest of the data. One of the most common methods used for anomaly detection is the isolation forest algorithm. This algorithm works by building a random forest of decision trees to isolate the anomalies from the rest of the data.

Unsupervised learning is widely used in a variety of fields, including anomaly detection [38], density estimation [39], and data compression [40]. Its ability to discover hidden structures in large datasets makes it a valuable tool for extracting useful insights and knowledge.

2.3.3 Semi-supervised Learning

Semi-supervised learning is a type of ML between supervised and unsupervised learning. It involves using labeled and unlabeled data to improve the accuracy of predictions or classifications. In semi-supervised learning, the goal is to leverage the available labeled data to make better predictions or classifications on the unlabeled data, using techniques such as self-training or co-training [41].

One of the main advantages of semi-supervised learning is that it can achieve higher accuracy than unsupervised learning with a smaller amount of labeled data [42]. This makes it an attractive option when labeled data is scarce or expensive. It is also a valuable tool for active learning, in which the goal is to selectively label the most informative data points to improve the model's performance [43].

There are several semi-supervised learning algorithms (see Figure 2.1), including self-training algorithms, co-training algorithms, and multi-view learning algorithms [41]. Self-training algorithms use a single model trained on labeled and unlabeled data [44]. Co-training algorithms involve using two or more models trained on different views of the data and used to label the unlabeled data iteratively [45]. Multi-view learning algorithms use multiple models trained on other data views and combined to make a final prediction or classification [46].

As in other types of learning, it is also essential to evaluate the model's performance on a separate test dataset to assess its generalization ability and to identify any overfitting or underfitting issues [47].

Semi-supervised learning is helpful for situations where labeled data is scarce or expensive. Its ability to leverage labeled and unlabeled data to improve the accuracy of predictions or classifications makes it a valuable tool for many real-world applications [48].

Figure 2.1 provides a conceptual summary of the machine learning methods commonly used for assessing human trafficking in social media. It illustrates the most common algorithms applied in previous research on trafficking-related content detection.

2.3.4 Naive Bayes

Naive Bayes uses sample data to estimate the posterior probability $P(y|x)$ for each class y and object x. Once we have these estimations, we can use them for classification. The foundation of Naive Bayes is the Bayes rule.

$$P(y|x) = P(y)P(x|y)/P(x) \tag{2.1}$$

Where:

- $P(y \mid x)$: Posterior probability of class y given observation x.
- $P(y)$: Prior probability of class y.
- $P(x \mid y)$: Likelihood of feature vector x given class y.
- $P(x)$: Evidence or marginal probability of observation x.

In addition to the presumption that, given the class, the attributes are conditionally independent. For attribute value data, this presumption grants

$$P(x|y) = \prod_{i=1}^{n} P(x_i|y) \tag{2.2}$$

where x_i is the value of the *ith* attribute in x, and n is the number of attributes.

$$P(x) = \prod_{i=1}^{k} P(c_i)P(x|c_i) \tag{2.3}$$

where c_i is the *ith* class and k is the number of classes.Therefore, by normalizing the numerators on the right side of the equation, the equation (2.1) can be computed.

2.3.5 Multinomial Naive Bayes

One of the Naive Bayes variations is used in text classification, where the data is usually represented by word vector counts. However, tf-idf vectors or bag of words are also known to perform well in practice, MultinomialNB implements the naive Bayes method for multinomially distributed data.

A set of parameter vectors characterizes the distribution, denoted as $\theta_y = (\theta_{y1}, ..., \theta_{yn})$, corresponding to each class y. Here, n represents the total number of features, which in the context of text classification equates to the vocabulary size. Furthermore, each θ_{yi} signifies the conditional probability $P(x_i|y)$, which is the likelihood of the occurrence of feature i in an instance classified under class y.

The parameters θ_y are ascertained through an enhanced variant of maximum likelihood estimation, namely, the method of relative frequency counting that incorporates smoothing techniques.

$$\hat{\theta}_{yi} = \frac{N_{yi} + \alpha}{N_y + \alpha n} \tag{2.4}$$

where $N_{yi} = \sum_{x \in T} x_i$ is the number of times feature i appears in a sample of class y in the training set T, and $N_y = \sum_{i=1}^{n} N_{yi}$

2.3.6 Support Vector Machine

Support Vector Machines (SVM) introduced by [49] are supervised learning models used for binary classification tasks. Their objective is to identify the best hyperplane that creates the widest possible separation between two classes in a high-dimensional space. This separation, called the margin, is measured as the distance from the hyperplane to the closest data points of each class—referred to as the support vectors.

The decision function of a linear SVM can be written as:

$$f(x) = \text{sign}(w^T x + b) \tag{2.5}$$

Where:

- $f(x)$: Predicted class label for input vector x
- w: Weight vector orthogonal to the separating hyperplane
- x: Input feature vector
- b: Bias term (intercept)
- $\text{sign}(\cdot)$: Sign function that returns $+1$ or -1 depending on the side of the hyperplane

SVMs are particularly effective in high-dimensional spaces and are robust to overfitting, especially in cases where the number of dimensions exceeds the number of samples. In real-world data, perfect separation is often not possible, so a soft-margin approach is employed, which allows for some classification errors.

The optimization problem for a soft-margin linear SVM is defined as follows:

$$
\begin{aligned}
\min_{w,b,\xi} \quad & \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}\xi_i \\
\text{subject to} \quad & y_i(w^T x_i + b) \geq 1 - \xi_i, \quad \xi_i \geq 0 \quad \forall i
\end{aligned}
\tag{2.6}
$$

Where:

- w: Weight vector defining the orientation of the hyperplane
- b: Bias term
- x_i: Feature vector of the i-th training sample
- $y_i \in \{-1, +1\}$: Label of the i-th sample
- ξ_i: Slack variable allowing misclassification
- C: Regularization parameter controlling the trade-off between maximizing the margin and minimizing classification error

In this formulation, ξ_i measures the degree of misclassification of the i-th sample. The parameter $C > 0$ determines how much misclassifications are penalized: a high value of C attempts to minimize misclassifications but may result in a smaller margin, while a smaller C allows more errors but promotes a wider margin.

In this work, we use the `SVC` class from the `scikit-learn` Python library, which implements this soft-margin formulation. The default kernel is linear unless otherwise specified.

2.3.7 Logistic Regression

Logistic Regression is a linear classification model that estimates the probability of a binary outcome using the logistic sigmoid function. It is widely used in binary classification tasks due to its simplicity, interpretability, and strong theoretical foundations.

The probability of an input vector x belonging to class 1 is given by:

$$
P(y = 1 \mid x) = \sigma(w^T x + b) = \frac{1}{1 + e^{-w^T x - b}} \tag{2.7}
$$

Where:

- $P(y = 1 \mid x)$: Probability that the input x belongs to class 1
- $\sigma(z)$: Logistic sigmoid function
- w: Weight vector
- b: Bias term
- x: Input feature vector

Model parameters are learned by maximizing the likelihood of the training data, or equivalently, by minimizing the cross-entropy loss:

$$E(w) = -\sum_{i=1}^{N} [t_i \log y_i + (1 - t_i) \log(1 - y_i)] \tag{2.8}$$

Where:

- t_i: True label for sample i (0 or 1)
- $y_i = \sigma(w^T x_i + b)$: Predicted probability

The gradient of the loss function is given by:

$$\nabla E(w) = \sum_{i=1}^{N} (y_i - t_i) x_i \tag{2.9}$$

This update rule resembles that of linear regression but uses probabilities instead of raw predictions. In practice, regularization (such as L2) is often added to prevent overfitting, especially when the data is linearly separable.

In this study, we use the `LogisticRegression` class from the `scikit-learn` library, which provides a robust implementation of logistic regression with regularization options to avoid overfitting.

2.3.8 Random Forest

Random Forest is an ensemble learning method, used for classification and regression tasks. It involves training a collection of decision trees at training time and taking the mode (majority vote) of some or all of the trees as output.

Each tree is trained on a different subset of the data using a procedure known as bootstrap aggregation or bagging. Also, at each split in a tree, only a random subset of features are considered. This adds randomness that brings diversity between the trees and reduces overfitting.

The final prediction for classification is given by:

$$\hat{y} = \text{mode}\left(T_1(x), T_2(x), \ldots, T_M(x)\right) \tag{2.10}$$

Where:

- $T_i(x)$: Prediction from the i-th decision tree
- M: Total number of trees in the forest
- $\text{mode}(\cdot)$: Majority vote among the trees

Random Forests demonstrate robustness to noise and maintain strong performance on high-dimensional data, even without feature scaling. Additionally, they exhibit lower sensitivity to hyperparameter variations compared to alternative machine learning models.

In this study, we use the `RandomForestClassifier` class from the `scikit-learn` library, which provides an efficient implementation of this algorithm.

2.4 Long Short Term Memory Networks

Recurrent neural networks (RNNs) can propagate historical information through a chain neural network structure. When processing sequential data, it will view the current input x_t and the output h_{t-1} of the previous time step in each time step. LSTM is a variant of recurrent neural networks. Since the traditional RNN is equivalent to a multi-layer feedforward neural network, the more significant the amount of historical data stored in RNN, the more layers. Training a large amount of text will cause the problem of gradient disappearance and loss of historical information. Therefore, the traditional RNN can retain a limited amount of historical information. The LSTM cell structure can solve the gradient disappearance problem of the RNN through the internal gating mechanism and better retain historical information. LSTM cell structure is shown in Figure 2.2.

Each time step of the LSTM architecture has a series of repeating modules. At each time step, the module's output is controlled by three gates, namely the input gate, output gate, and forget gate. f_t , i_t and O_t are denoted as forget gate, input gate and output gate at time t respectively, U_t is the temporary cell state of the memory unit, and C_t is the final cell state of the current memory unit. The internal of LSTM is mainly divided into three stages: the first stage is to control the historical information stored in the memory unit and selectively forget the contents of the memory unit passed in by the previous node through the forget gate. The second stage is selective memory, which uses the input gate to selectively memorize the current temporary cell state's information and some critical information. The sum of the results of the first two stages is the current cell state of the memory unit, which is used to transfer the cell state to the next memory unit. The third stage is the output. The output gate controls the output of the LSTM, and the final production is recorded as h_t. Let $X = x_1, x_2, ..., x_n$ be the input text matrix, x_t is the word vector at time step t, with dimension d, The hidden vector h_t (the size is H) at the time step t is updated as follows.

$$i_t = \sigma(W_i X_t + U_i h_{t-1} + b_i) \tag{2.11}$$

$$f_t = \sigma(W_f X_t + U_f h_{t-1} + b_f) \tag{2.12}$$

$$o_t = \sigma(W_o X_t + U_o h_{t-1} + b_o) \tag{2.13}$$

$$u_t = \tanh(W_u X_t + U_u h_{t-1} + b_u) \tag{2.14}$$

$$C_t = i_t \odot u_t + f_t \odot C_{t-1} \tag{2.15}$$

$$h_t = O_t \odot \tanh(C_t) \tag{2.16}$$

W_x, U_x and b_x ($x \in i, o, f, u$) are model parameters, σ is the sigmoid activation function, LSTM makes the information calculated in the subsequent moments pass by forgetting the old information and storage of new information in the state of the memory unit. The useless information will be discarded, and the hidden layer state h_t will be output at each time step. Among these, h_t contains the historical information of the previous $t-1$ times; finally, we can get the hidden layer state sequence $h_1, h_2, ..., h_n$ with the same length as the input text.

In our LSTM implementation, the sigmoid function $\sigma(x) = \frac{1}{1+e^{-x}}$ was used for the input, forget, and output gates, while the hyperbolic tangent function $\tanh(x)$ was used for the candidate memory state. These activation functions are standard in LSTM architectures

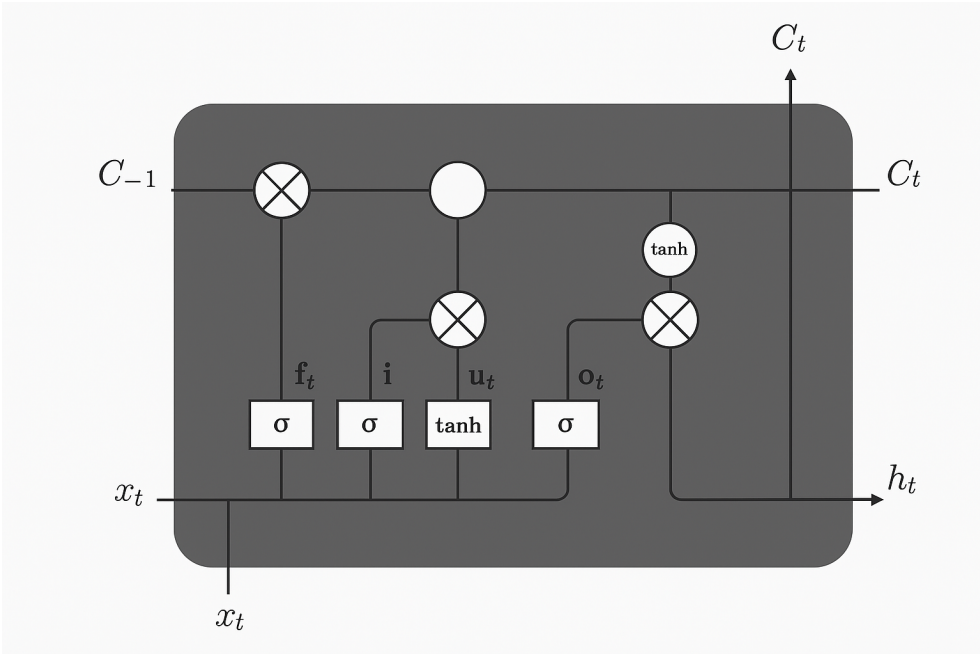

Figure 2.2: LSTM cell structure

due to their proven effectiveness in controlling the flow of information through gating mechanisms. Empirically, we observed that the sigmoid function helped maintain stable gradient flow and worked well for capturing the sequential dependencies in tweet data.

This behavior aligns with the default configuration of TensorFlow's `LSTM` layer, which uses `activation='tanh'` for the candidate cell state and `recurrent_activation='sigmoid'` for the gating functions.

2.5 Metrics

2.5.1 Accuracy

These are primary metrics to evaluate the quality of a classification model. Let TP, FP, TN, and FN denote true positive, false positive, true negative, and false negative, respectively. The classification Accuracy are defined in Eq. 2.17

$$\textbf{Accuracy} = \frac{TP + TF}{N} \tag{2.17}$$

where N is the total number of samples

2.5.2 Precision

Precision is a fundamental metric in the field of information retrieval and classification, particularly in the context of binary classification tasks. It quantifies the accuracy of optimistic predictions made by a classifier. Mathematically, precision is defined as the

ratio of true positive predictions to the total number of positive predictions as in Eq. 2.18, including both true and false positives. Precision measures the classifier's ability to identify positive instances while minimizing false positives correctly. A higher precision indicates a lower rate of false alarms and greater confidence in positive predictions.

$$\textbf{Precision} = \frac{TP}{TP + FP} \tag{2.18}$$

2.5.3 Recall

Recall, also known as sensitivity or true positive rate, is another vital metric for evaluating the performance of a classifier, especially in scenarios where identifying all positive instances is of paramount importance, such as medical diagnosis or fraud detection. Recall measures the ratio of correctly predicted positive instances to the total number of actual positive cases in the dataset as in Eq. 2.19, which includes both true positives and false negatives. It represents the classifier's capability to capture all relevant positive instances. A higher recall indicates a lower likelihood of missing positive cases and a more comprehensive coverage of positive instances.

$$\textbf{Recall} = \frac{TP}{TP + FN} \tag{2.19}$$

2.5.4 F1-score

The F1 score is a composite metric that balances precision and recall. It is beneficial when there is a trade-off between these two metrics, as increasing precision often decreases recall and vice versa. The F1 score is the harmonic mean of the precision and recall, as in Eq. **??**. An F1 score reaches its best value at 1 (perfect precision and recall) and worst at 0.

$$\textbf{F1-score} = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{2.20}$$

2.6 Text representation

Text representation, a fundamental concept in natural language processing (NLP), is the cornerstone for various text-based applications, including information retrieval, text classification, sentiment analysis, and machine translation. Text representation aims to convert unstructured textual data into structured, numerical forms that algorithms and models can effectively process and analyze text. This transformation is essential because computers and machine learning models primarily operate on numerical data. Text representation techniques bridge the semantic gap between human language and computational systems, enabling the extraction of meaningful insights and patterns from vast textual corpora.

2.6.1 Bag of Words (BoW)

The Bag of Words (BoW) model is foundational in natural language processing (NLP) and text analysis. It is a simple yet powerful approach for representing and processing

textual data. The BoW model operates under the assumption that the order of words in a document is irrelevant, focusing solely on the frequency or presence of individual words. In this model, a document is treated as an unordered collection of words, and each word is considered an independent feature. Due to its simplicity and effectiveness, BoW has found extensive applications in various NLP tasks, such as text classification, sentiment analysis, information retrieval, and document clustering.

2.6.2 Word Embeddings

Word embeddings, also known as word vectors or word representations, have gained immense popularity in recent years as a more context-aware approach to text representation. Word embeddings leverage neural networks to learn dense vector representations of words that capture semantic relationships and contextual information. These embeddings transform words into high-dimensional vectors, where similar words are closer in vector space. Pre-trained word embeddings, such as Word2Vec [50], GloVe [51], and FastText [52], have demonstrated remarkable success in various NLP tasks. By encoding semantic meaning and context, word embeddings enable models to better capture the nuances of language and improve downstream applications' performance.

2.6.3 Document Embeddings

In addition to word-level representations, document embeddings aim to capture the overall context and semantics of entire documents. Methods like Doc2Vec [53] or ada-002 [54] from OpenAI and paragraph embeddings employ neural networks to generate fixed-length vector representations for documents. These embeddings encode the words within a document and the relationships between them. Document embeddings are particularly valuable in applications such as information retrieval, where the goal is to retrieve documents that are semantically similar to a given query. By capturing document-level semantics, these embeddings enhance the accuracy and relevance of retrieval systems.

Chapter 3

State of the Art

3.1 Text classification

Text classification, a fundamental task in natural language processing (NLP), plays a pivotal role in numerous applications, including sentiment analysis, topic modeling, spam detection, and detection of messages related to human trafficking. Text classification has witnessed significant advancements in recent years, driven by innovations in machine learning and deep learning techniques. This section provides an overview of the state-of-the-art methods, strategies, and trends in text classification, highlighting key developments that have shaped the landscape of this domain.

The work of *Pang et al.*[55] does a binary text classification on IMDb movie reviews. For this work, they used three different ML algorithms: Support Vector Machine, Naive Bayes, and Maximum Entropy. In addition, they tested the use of unigrams and bigrams, including part of speech and position. In addition, they tested using word frequency or presence alone. The authors found that the presence of unigrams is the most effective way, but also, in combination with bigrams, it further improves the performance of the classifiers. Finally, they found that it is much more effective to use word presence compared to word frequency. The best-performing algorithm in their work was the Support Vector Machine, with an accuracy of 82.9%. One of the limitations mentioned by the authors is that the use of Bag of Words on several occasions can confuse one class with another since, many times, the text itself can have words that dictate the opposite class to the one it belongs to. However, if a human does it, it is easy for him to find the correct class. What they suggest is to use some form of discourse analysis.

On the one hand, Alec Gocite [56] proposes using Naive Bayes, Support Vector Machine, and Maximum Entropy for sentiment analysis with distant supervision using Twitter data. In this work, they use feature reduction tagging usernames and links as USERNAME and URL, respectively. They also reduce words with repeated letters, for example, *helloooooo* to *hello*. The ML algorithm with the highest accuracy was Maximum Entropy with 83%, using unigrams and bigrams in conjunction. Finally, the authors mention that they did not achieve state-of-the-art results. However, this may be because they are classifying using several classes and that using them in a specific domain may increase the performance. In addition, they mention that consideration should be given to using these classifiers in

languages other than English. Finally, they say that it would be essential to see the effects of including emoticons as part of the input data since they eliminated them, and they can give more context to the text for better classification.

On the other hand, *Jiang et al.*[57] propose using Naive Bayes, Support Vector Machine, and Logistic Regression for the binary classification of text obtained from Twitter. The classification you do is a sentiment analysis, but they also add something new, and that is the use of subjectivity and objectivity. They label with TextBlob each tweet if it is subjective or objective by defining a threshold. Then, they eliminate the tweets labeled as subjective and the objective ones they use to train the algorithms mentioned above. This approach significantly improves the performance of the Machine Learning models. The best-performing algorithm is Logistic Regression, with 81.7% accuracy using unigrams and bigrams together.

3.2 Human trafficking detection in social media

3.2.1 Supervised Approaches

Table 3.1 summarizes the works using supervised algorithms. Supervised ML algorithms are commonly used for detecting human trafficking, as they require labeled data to learn from. However, obtaining labeled data for human trafficking is challenging, as it involves sensitive information and can potentially put individuals at risk. Despite this challenge, several approaches have been taken to obtain labeled data, including using pre-existing data or manually labeling data pulled from social media platforms like Twitter.

The data extraction typically involves using web scraping techniques or social network APIs, followed by preprocessing steps such as removing links and non-relevant characters such as emoticons and emojis. The labeled data is then fed into the supervised ML algorithm for further analysis.

Once the data is ready, it is commonly split into training and testing subsets, or in some cases, into three subsets: training, validation, and testing. The proportion of the split varies depending on the researcher, but a standard split is 80% for training and 20% for testing.

The algorithms most commonly used for supervised learning in human trafficking detection include Support Vector Machine [22, 58, 20, 59, 18, 60, 61, 62, 63, 19], Logistics Regression [60, 62, 63, 19], Random Forest [59, 60, 62, 63, 19], Gaussian Naive Bayes, and Artificial Neural Networks such as the multi-layer perceptron. These algorithms form the basis for supervised learning in classification, detection, and regression tasks.

To evaluate the performance of the classifiers, metrics such as Precision, Recall, F1-score, Accuracy, and AUC are commonly used. These metrics provide insights into the algorithm's effectiveness in detecting human trafficking and can guide future improvements in the methodology. Overall, supervised learning approaches have proven to be effective in detecting human trafficking despite the challenges of obtaining labeled data.

Finally, supervised ML algorithms have become popular for detecting human trafficking. Despite the challenges in obtaining labeled data, various approaches have been taken to acquire it, such as using pre-existing datasets and manually labeling data extracted from social media platforms.

3.2.2 Unsupervised Approach

Table 3.2 summarizes the works using unsupervised algorithms. Unsupervised techniques have also been employed to detect human trafficking, as they do not require labeled data to train the algorithms. Clustering algorithms are commonly used to group data together based on similarities in their features. One of the most used clustering algorithms in human trafficking detection is k-means clustering.

Once the unsupervised techniques have been applied, the researcher can further refine the results by manually examining the data points flagged as potential instances of human trafficking. This manual approach can eliminate false positives and increase the accuracy of the results.

Lastly, while supervised learning algorithms are commonly used in human trafficking detection, unsupervised techniques such as clustering [64, 65] and anomaly detection [66, 67] can also be practical tools in identifying instances of human trafficking. These techniques can be used when labeled data is unavailable and can be used in conjunction with supervised learning algorithms to improve their accuracy.

3.2.3 Semi-supervised Approach

Table 3.3 summarizes the works using semi-supervised techniques that combine unsupervised and supervised methods. They have shown promising results in human trafficking detection. One such hybrid approach is to use unsupervised techniques for word embedding and supervised techniques for the detection task.

Word embedding is a technique used to represent words in a vector space, where words with similar meanings are closer to each other. Word embedding techniques such as a bag of words, word2vec, TF-IDF, FastText, and Skip-grams are commonly used. In the context of human trafficking detection, word embeddings can be used to represent words and phrases that are commonly associated with human trafficking, such as "sex trafficking" or "forced labor."

Once the word embeddings are generated, they can be used as input to a supervised learning algorithm, such as a neural network or a support vector machine. The supervised algorithm is trained on labeled data to learn the relationship between the word embeddings and the presence of human trafficking activity. The labeled data can be obtained using earlier methods, such as manual labeling or web scraping.

The advantage of using a semi-supervised approach is that it leverages the strengths of both unsupervised and supervised techniques. Unsupervised techniques can generate high-quality word embeddings, while supervised techniques can be used to learn the relationship between the embeddings and human trafficking activity. This approach can also be practical when labeled data is limited, as it can augment the available labeled data with word embeddings generated from a larger, unlabeled dataset.

Moreover, it is used for a small amount of labeled data to train a model and then the trained model to label a more significant amount of unlabeled data. The labeled and unlabeled data can be trained in a new model. This iterative process can be repeated until the model's accuracy is satisfactory.

Indeed, semi-supervised techniques that combine unsupervised techniques for word embedding and supervised techniques for human trafficking detection have shown promise in

detecting human trafficking activity. This approach can be practical when labeled data is limited and can leverage the strengths of both unsupervised and supervised techniques. Moreover, semi-supervised techniques can be helpful also.

3.2.4 Datasets

Data extraction

For the extraction of data, what is used are public web pages where there may be indications of human trafficking, such as adult service pages and pornographic pages where sex work is offered. However, in reality, they are human trafficking networks. Another common avenue for recruiters is social networks. They can contact potential victims more easily by posing as friends, acquaintances, or job recruiters. Micro-blogging platforms are the most used by traffickers because they allow for more significant interaction between strangers, and since they share day-to-day thoughts, they can more easily identify potential victims and form a rapport.

One of the most commonly used sites for data extraction using web scraping techniques is Backpage.com. Backpage.com was a classified ads website founded in 2004 by the alternative newspaper chain New Times Inc./New Times Media. It allowed users to post ads in categories such as personals, automotive, rentals, jobs, and adult services. The latter type was used by human trafficking rings to recruit potential victims for their network. For this same reason, it is widely used to extract advertisements related to human trafficking as used in [63].

Other sites used in the work of [59] for data mining are news and advertisements in general. These are known to contain job offers that are methods used by traffickers to recruit new victims. Likewise, YELP reviews are also used to detect places where sexual services are provided based on keywords such as Massage, Beauty and spa, Health and Medical, Nail Salons, Day Spas, Reflexology, Physical Therapy, Skin Care, and Tui Na.

Finally, social media contain a wealth of data that can be used to detect this problem. One of the most widely used data extraction services is Twitter [68, 22, 58, 20, 18, 66, 19, 64], which has a freely available API that facilitates the retrieval of tweets using queries and keywords, without using web scraping techniques. Another of the most used social networks is Reddit, which has a free API that facilitates the extraction of tweets using queries and keywords.

Data Labeling

Due to the complexity of the problem of human trafficking, it is very complicated to find publicly labeled data since it may contain sensitive information and data of persons who may or may not be involved, such as phone numbers, addresses, names, etc. Therefore, to request this data, it is usually necessary to ask the author to explain that it is for research and describe the area and what you plan to do with it.

On the other hand, labeling data depends on having a person review each piece of data by hand and assign a label, whether or not it is related to human trafficking. This process is highly dependent on the person's judgment, and there may be a bias in assigning the label. Therefore, this task usually requires a person with expertise in human trafficking to

review and label the data. Their experience allows us to minimize bias and ensure that the data are correctly labeled.

Another problem is that manual review is required, and the need for an expert in the field to obtain labeled data limits us with the number of data. There are not many people with expertise in the area of human trafficking who are willing to review thousands or millions of pieces of data and tag them. Also, not all universities, research centers, or research groups have people with suitable characteristics to perform this task.

In human trafficking, there is a public dataset (upon request to the authors) with labeled data that is used to train ML models to detect advertisements with indications of human trafficking. This data is called Trafficking-10k [62] and comprises 10 thousand ads and seven different tags: Certainly no, Likely no, Weakly no, Unsure, Weakly yes, Likely yes, Certainly yes.

3.2.5 Pre-processing

Several preprocessing steps are commonly followed by researchers when text is used to classify it. These steps aim to clean and transform the raw tweet data into a form suitable for further analysis. Some of the most commonly used preprocessing steps are:

Tokenization

Tokenization is the process of breaking a text into individual words or tokens. In tweets, tokenization can be challenging due to the presence of emoticons, hashtags, and mentions. Therefore, specialized tokenization techniques. The works of [19, 20, 58] tokenize the text from the tweets to have a better understanding of the model and gain performance.

Stopword Removal

Stopwords are common words that do not carry much meaning, such as “the,” “a,” and “an.” Removing stopwords can reduce the dimensionality of the data and improve the efficiency of subsequent analysis steps. However, the effectiveness of stopword removal in text has been debated in the literature, with some studies suggesting that it may harm the performance of classification models. In the case of the works [22, 60, 59] extract the data from social media platforms, so they remove stopwords less dimensionality to the model, and gain meaningful data for the model.

Stemming/Lemmatization

Stemming and lemmatization are techniques for reducing words to their root form. This can help to decrease the sparsity of the data and improve the accuracy of subsequent analysis steps. However, stemming can also result in the loss of information, and lemmatization can be computationally expensive. Therefore, the choice of stemming/lemmatization technique may depend on the specific task and dataset.

Removing URLs, emojis, mentions, and hashtags

Tweets often contain URLs, mentions, emojis, and hashtags, which can be irrelevant or misleading for classification tasks. Therefore, these elements are usually removed before analysis. The works [18, 69] extract the data from Twitter, so they come with too much poor information such us URLs, emoticons, mentions, and hashtags that are affecting the performance of the ML models.

Spell correction

Text from social media often contains misspellings and abbreviations, making it challenging to analyze the data accurately. Therefore, spell correction techniques, such as the use of a spell-checker, can be used to improve the accuracy of subsequent analysis steps.

Normalization

Text normalization included converting all characters to lowercase, removing punctuation, and reducing letter repetitions (e.g., holaaaa → holaa). Additionally, hashtags, usernames, and URLs were replaced with special tokens (e.g., ___HASHTAG_word, ___USER_, ___URL_). Although traditional numeric normalization was not applicable to text, these linguistic normalization steps helped standardize the inputs for embedding generation.

Table 3.1: Supervised Machine Learning Approaches to Address Human Trafficking on Social Media

Authors	**Input Feature**	**Classes**	**Model**	**Dataset**
Vieira et al.[68]	Tweets: 13.6k Images: 11.1k	NA	OCR	Twitter Police Data
Shishira et al.[22]	Tweets: 155.1k Images:89k	(suspicious, not suspicious), (man, woman), (under14, over14)	SVM VGG16	Twitter IMDB-Wiki Other DS
Hernández and Granizo [58]	Tweets: 55.1k Images: NA	(suspicious, not suspicious), (man, woman), (under14, over14)	SVM	Twitter
Wiriyakun et al.[9]	Ads: 10k	(trafficking, not trafficking)	LIME, NB, DT, ANN	Trafficking-10k
Granizo et al.[20]	Tweets: 100k Images: 820	(suspicious, not suspicious), (man, woman), (under14, over14)	SVM NB CNN	Twitter
Diaz et al.[59]	Reviews: 456k	(illegal activity, not illegal activity)	SVM, NB, RF, ANN	Yelp reviews Rubmaps DS
Hernandez[18]	Tweets: 55.1k	(suspicious, not suspicious)	SVM NB	Twitter
Goist et al.[60]	Text: 178 country reports	(sending relationship, receiving relationship)	LR, SVM, NB, RF	TIP reports
Zhu et al.[61]	Ads: 10k	(high-risk, low-risk)	KenLM, LR, SVM	Trafficking-10k
Tong et al. [62]	Ads: 10k	(certainly no, likely no, weakly no, unsure, weakly yes, likely yes, certainly yes)	RF, LR, SVM, HTDN	Trafficking-10k

Table 3.2: Unsupervised Machine Learning Approaches to Address Human Trafficking on Social Media

Authors	**Input Feature**	**Classes**	**Model**	**Dataset**
Li et al.[65]	Ads: 10M	(positive, negative)	HDBSCAN	Backpage
Burbano and Hernández[66]	Tweets: 167k	NA	NLP	Twitter
Lee et al.[64]	Tweets: +18M Ads: 10k	(certainly no, likely no, weakly no, unsure, weakly yes, likely yes, certainly yes)	InfoShield	Twitter Trafficking-10k
Shelke et al.[67]	Images: +3M	NA	K-Means	FaceScrub[1] VGG[2] FPA[3]

Table 3.3: Semi-supervised Machine Learning Approaches to Address Human Trafficking on Social Media

Authors	**Input Feature**	**Classes**	**Model**	**Dataset**
Alvari et al. [63]	Ads: 20k	(positive, negative)	S3VM-R SVM KNN NB LR Adaboost RF	Backpage
Wang et al. [70]	Ads: 20k Emojis	(certainly no, likely no, weakly no, unsure, weakly yes, likely yes, certainly yes)	ORNN Skip-gram	Trafficking-10k
Alvari et al. [71]	Ads: 20k	NA	K-Means KNN RBF	Backpage
Kejriwal et al. [72]	Text	NA	Flexible and adaptive generation of indicators	Web Resources
Ramchandani et al. [73]	Posts: 13.5M	(suspicious, not suspicious)	CBOW	Deep Web
Burbano and Hernández [19]	News: 814	(related to human trafficking, not related)	SVM LR RF DT Perceptron SGD KNN	News
Tudis et al.[69]	Tweets: 122k	(suspicious, not suspicious)	RF BoW	Twitter

Chapter 4

Methodology

4.1 Data Collection and Extraction

The process of constructing this dataset began with the extraction of tweets from X (before Twitter) vast repository of user-generated content. Leveraging the Twitter API, we systematically gathered tweets that exhibited potential relevance to the issue of human trafficking. We took great care to ensure that the collected tweets spanned a diverse range of topics, discussions, and geographical locations, thereby capturing a holistic representation of the subject matter on the platform. Data collection was carried out using Python's twarc1 module, which utilizes pagination to overcome the limitations on tweet numbers set by Twitter for queries. All the tweets obtained were in Spanish. The query included hashtags and keywords related to human trafficking and child pornography. The tweets were collected from June 2022 to June 2023, resulting in 16995 raw tweets.

Table 4.1 shows the hashtags and keywords with their meaning used in the query for the tweet extraction. The query used is the following:

query ="(#cp OR #loli OR #lolita OR #joven (#fresca OR #dulce OR #nueva) OR cp (niña OR lolita OR loli OR joven OR nueva) OR peladitas OR niña (sexy)) lang:es -is:retweet -is:nullcast'

4.1.1 Manually annotated

We initiated a labor-intensive manual labeling process to establish ground truth annotations for the ML and LSTM model's training and evaluation. Human annotators meticulously reviewed each tweet with expertise in the field to determine its category.

In the labeling process, we applied specific rules to categorize tweets into three distinct labels: 'Suspected' (related to human trafficking or illegal sexual services), 'Not Suspected' (pertaining to the sexual domain but not suspected of illegal activities), and 'Unrelated' (tweets unrelated to the specified topics). Below are the rules and criteria used for each label:

Table 4.1: Description of the hashtags and keywords used for the tweet extraction

Hashtag and keywords	Meaning
#cp	child pornography
#joven	young
#loli	young girl
#nueva	new
#fresca	fresh
#dulce	cute
lolita	young girl
niña	girl
peladitas	young girls

Not Suspected (Sexual Domain) - Label = 0:

- Promotion of personal sexual services or the sale of sexual content (e.g., Onlyfans, companionship, escorts), where it is evident that the individual is of legal age and promoting their services voluntarily.
- Tweets related to sexual roleplay on Twitter.
- Reports or mentions of potential sexual crimes.
- References to other individuals regarding the possession, viewing, or promotion of sexual content, provided that there is no suspicion of illegal activity.
- Sexualized comments.

Suspected (Human Trafficking or Illegal Sexual Services) - Label = 1:

- Tweets that explicitly detail potential instances of human trafficking or forced sexual services, such as the promotion of sexual services involving minors or individuals promoting sexual services provided by others. (Note: Self-promotion by adults does not fall under this label; it falls under 'Not Suspected.')
- Tweets indicating the creation, possession, or distribution of illegal sexual content.
- Tweets suggesting an intention to abduct or manipulate underage women for illegal sexual activities.
- Tweets promoting groups/chats on platforms like WhatsApp, Telegram, Reddit, or others where the distribution of child pornography or the offering of sexual services involving minors is evident.
- Tweets in which a minor promotes their own sexual services or content.

Unrelated - Label = 2:

- Tweets that use the specified keywords in an unrelated context. For instance, the use of 'CP' as an abbreviation for postal codes or the criminal code rather than child pornography.
- Tweets that mention users with keywords in their usernames. For example, tweets in response to users with 'CP' in their usernames.
- Tweets are addressing topics entirely unrelated to human trafficking, legal or illegal sexual services, and child pornography.
- Promoted tweets.
- Spam content.

This labeling process was meticulously applied to ensure the accurate categorization of tweets into these three distinct labels, thereby enabling the subsequent development and evaluation of our LSTM-based tweet classification model.

4.2 Proposed Methodology

For this study, our proposed approach involves utilizing machine learning algorithms to establish a foundational framework for the new dataset. Subsequently, we aim to compare the outcomes of these algorithms with models employing Long Short-Term Memory (LSTM) networks. In this context, we present two distinct approaches, one considering the inclusion of emojis and another without them.

4.3 Data Set Description

The foundation of any machine learning model's success lies in the quality and relevance of the dataset used for training and evaluation. In this study, we present a meticulously curated dataset tailored to the problem of detecting tweets related to human trafficking on social media platforms. This dataset is an essential resource for training and validating our proposed LSTM-based tweet classification model we propose. To ensure the dataset's reliability and accuracy, we employed a manual collection and labeling process, which involved the extraction of tweets from Twitter using the Twitter API. The dataset comprises tweets categorized into two distinct classes: 0, representing sexually explicit content unrelated to human trafficking, and 1, signifying tweets explicitly linked to human trafficking.

The dataset comprises a total of 6,312 tweets, of which 5,357 belong to category 0 (tweets of a sexual nature), while 955 are categorized as belonging to category 1 (possibly related to human trafficking), as illustrated in Figure 4.1. The dataset exhibits a noticeable imbalance. This imbalance is primarily rooted in the intricate nature of the problem, involving activities that perpetrators carry out stealthily on social media. This challenge underscores the necessity of developing machine learning models capable of accurately discerning the category with fewer available instances. In the model evaluation process, it

is imperative to consider that relying solely on the accuracy metric may prove insufficient, given the dataset's imbalance. Therefore, additional metrics that adequately address this imbalance will be employed for a more comprehensive assessment. These metrics include Precision, Recall, and F1-Score. These metrics will be utilized both during the training and validation phases of the models, aiming to obtain a more holistic evaluation of their performance and their ability to address the inherent challenge of category imbalance.

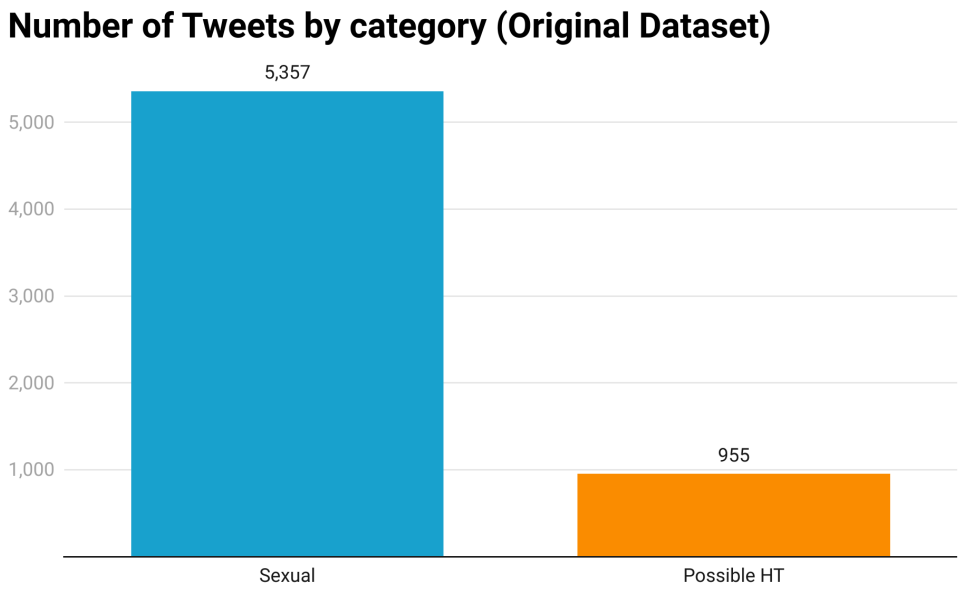

Figure 4.1: Dataset distribution by class without data augmentation

4.4 Pre-processing

In our preprocessing workflow, we utilize two distinct preprocessing methodologies, one designed for Machine Learning algorithms and the other for Deep Learning algorithms. We will now provide a detailed description of each:

4.4.1 Stop-words

We used Spanish vocabulary from the NLTK library to eliminate stop words for this task. In the context of Spanish text processing, examples would include terms such as 'el,' 'la,' 'de,' 'en,' 'en,' 'y', 'o', 'con,' 'por,' 'para,' 'us,' 'una,' among others. Identifying and eliminating these words in the early stages of analysis can result in more compact and meaningful text representations.

4.4.2 Repetitions

Sometimes, there are misspelled words with repetition letters, so we must deal with that. We reduce the number of repetitive letters in the form:

holaaaa → holaa

4.4.3 Steamming/Lemmantization

Stemming and lemmatization are techniques for reducing words to their root form. This can help decrease the data's sparsity and improve the accuracy of subsequent analysis steps. For this work, we use the library Spacy for lemmatization. It has a corpus in Spanish, which is required for the dataset.

4.4.4 Hashtags

In the case of Hashtags, we tagged the hashtags in the following form:

#lolita → ___HASHTAG__lolita

4.4.5 URL's

In the case of URLs, we replace all the URLs with the tag: **___URL__**

4.4.6 User names

To handle the usernames, we tagged them in the following form:

@ElonMusk → ___USER__elonmusk

4.4.7 N-grams

N-grams are contiguous sequences of 'n' items, typically words or characters, extracted from a larger text or dataset. They are a fundamental concept in natural language processing and text analysis. For example, in the case of uni-grams or 1-grams, each word in a sentence represents a distinct uni-gram. Bi-grams (2-grams) consist of pairs of consecutive words, while tri-grams (3-grams) consist of triples, and so on. For this work, we use 1-gram, 2-gram, and 3-grams together to give more features to the model.

4.4.8 Emoticons and Emojis

In handling emoticons, a pivotal procedure was undertaken, which involved the translation of emoticons identified within the text into their respective emoji representations, each accompanied by its corresponding Unicode Identifier. Subsequently, the analysis incorporated these emojis, and their respective subgroups were included as features. This augmentation serves the purpose of semantically contextualizing the intended meaning conveyed by each emoji. In the ensuing section, Table 4.2 provides a comprehensive tabulation of this translation process.

Table 4.2: The corresponding transformation between emoticons, emojis, and subgroup (meaning)

Emoticons	Emoji	Unicode	Sub-group
:-), :), (:, (-:	😁	U+1F601	face-smiling
:-D, :D, X-D, XD, xD, xd	🤣	U+1F923	face-smiling
<3, :*	❤️	U+2764	heart
;-), ;), ;-D, ;D, (; , (-;	😉	U+1F609	face-smiling
:-(, :(,):,)-:	☹️	U+2639	face-concerned
:,(, :(, :'(, :((	😢	U+1F622	face-concerned

4.5 Data Augmentation

This work used data augmentation to combat the imbalance of the data obtained through Twitter. For this task, the OpenAI API and the gpt-3.5-turbo-0125 model were used to generate new samples with synonyms and similar words based on the tweets from the category with the least number of examples, category 1. For this task, a dynamic prompt in Spanish was designed, which included a tweet as an example, taking each one from class 1. Ultimately, the LLM generated three new examples from the original tweet. Here is the prompt template and an example of response:

Generate three tweets based on an example tweet following these rules:

1. Try to use synonyms and similar words but maintain the intention of the original tweet.
2. Do not censor any words.
3. Do not remove hashtags or emojis/emoticons.
4. If no example tweet is provided, DO NOT GENERATE ANYTHING; return an empty JSON.
5. The output format should be a JSON with the following structure '{'1': 'tweet1', '2': 'tweet2', '3': 'tweet3'}'

Here is the example tweet: {Buenos días baby Me encuentro ya #disponible #sabrosa #lolita #cdmx 552***}

Response:

{'1': 'Saludos amorcito, ya estoy #disponible #deliciosa #nena #cdmx 552***',
'2': '¡Hola bebé! Aquí estoy #disponible #sexy #jovencita #cdmx 552***',
'3': '¡Buen día mi vida! Disponible para ti #sabrosa #dulce #cdmx 552***'}

After generating new samples for the imbalance class, the distribution of the tweets by classes with data augmentation is in Figure 4.2. The dataset is still imbalanced, but we reduced the class gap.

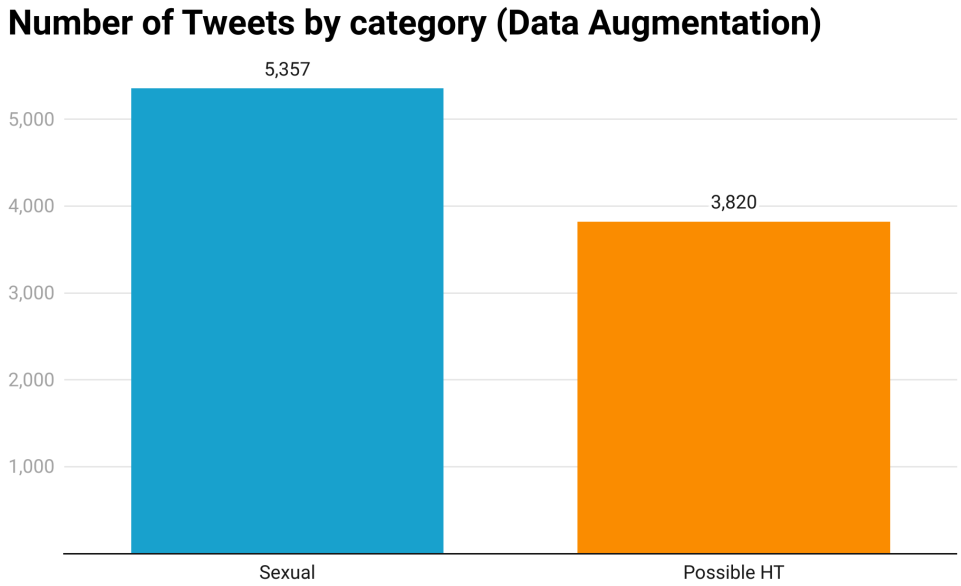

Figure 4.2: Dataset distribution by class with data augmentation

4.6 Experiments

The main objective of this work is to study the effect of using LSTM with this new dataset for binary text classification in the field of human trafficking messages. Nevertheless, we want to perform a bunch of experiments and analyze the performance of each of them.

Moreover, the first case of this study is to establish a foundational framework for the new dataset. In this case, we experiment using machine learning algorithms to classify sexual tweets (class 0) or suspicious tweets related to HT (class 1). This experiment uses four machine learning algorithms named Gaussian Naive Bayes, Logistic Regression, Random Forest, Support Vector Machine, and Maxent.

Another aspect that we evaluate is the effects of emojis on text classification. For this evaluation, we conduct each model training with and without emojis. It includes the ML algorithms and LSTM-based models. For the ML algorithms, we use BoW as text representation, and as a last feature, we count what kind of emojis are present in the tweet.

For the LSTM-based model, we used word embedding, specifically Word2Vec. The advantage of Word2Vec is the use of the pre-trained models. For instance, we use for text a Spanish Word2Vec pre-trained model [74], and for emojis a Word2Vec pre-trained model for emojis called Emoji2Vec[75].

4.6.1 One stream architecture

This proposal will employ a single stream containing embedded text representations without considering emojis.

4.6.2 Two stream architecture

We employ a dual-stream approach with two separate input channels in this model. Each of these channels is associated with its embedding layer. The primary distinction between these channels lies in the fact that, through one stream, only the embedded representations of text are forwarded. In contrast, the embedded representations of emojis are transmitted through the other stream.

4.7 Hardware and Software

All experiments were conducted on a Gamer Laptop Asus ROG Strix G16 (2023). The main configuration used for training LSTM models was as follows:

- CPU: Intel Core i7 13650HX
- GPU: NVIDIA GeForce RTX 4060 8GB vRAM Laptop GPU
- RAM: 64 GB DDR5
- OS: Windows 11 Home (x64)
- Frameworks: Python, TensorFlow, and Scikit-learn, NLTK

This configuration allowed for efficient training of deep learning models, especially when processing large batches of tweet embeddings.

Chapter 5

Results and Discussion

In this chapter, the results obtained during the experimentation will be exposed. Part of the experiments has been performed with ML algorithms and the other part with algorithms based on LSTM.

5.1 Analysis approach

The planned analysis involves the utilization of four metrics, namely accuracy, precision, recall, and F1 score. During the discussion, we will delve into the factors that impact these results, such as the usage of distinct models, different methods of text representation, and whether or not emojis were included. Additionally, we will explore whether data augmentation improved the models' performance.

5.2 Machine Learning Algorithms

The dataset consists of 6,312 manually annotated tweets in Spanish, collected using targeted queries designed to capture language and hashtags associated with human trafficking. While the dataset size may appear modest, the focused sampling strategy based on domain-specific keywords ensures that the examples are highly relevant. Manual labeling by trained annotators adds quality to the data. Nonetheless, the study acknowledges that future research with larger and more diverse samples would further improve generalizability.

The dataset was divided into three parts: 70% for training and 15% for each test set conducted for each algorithm.

The Bag of Words method was employed for the machine learning algorithms for text representation.

In general terms, there isn't much difference between the results of machine learning algorithms that include emojis as features (refer to Table 5.1) and those that do not include emojis as features (refer to Table 5.2). This may be attributed to the text representation used for these algorithms, as it fails to effectively capture the intention behind the emojis in the text. Consequently, we will focus on the results where emojis were included as features.

If we examine Table 5.1, we observe that the best-performing model in terms of accuracy during the training phase is Random Forest (RF) with 98.8%, followed by Logistic Regression (LR) with 99.1% accuracy. It appears that these algorithms can adapt well to the training set. However, a slight difference can be noticed between their accuracy in test 1 and test 2. For example, Logistic Regression and Maxent have the highest accuracy in the validation set, respectively. This suggests that they are better at generalizing. Nevertheless, when examining other metrics, it becomes apparent that all machine learning algorithms used in this study struggle to detect the HT class.

Table 5.1: Summary of accuracy from ML models with emoji features

Model	Train	Test 1	Test 2	Train Time (s)	Eval Time (s)
Naive Bayes	0.64	0.84	0.83	83.51	157.98
Support Vector Machine	0.98	0.94	0.92	83.19	161.20
Logistic Regression	0.99	**0.94**	0.93	**82.58**	160.56
Random Forest	**0.99**	0.93	0.91	92.06	**157.60**
Maxent	0.99	0.92	**0.94**		

Table 5.2: Accuracy from ML models without emoji features

Model	Train Acc	Test 1	Test 2	Train Time (s)	EvalTime (s)
Naive Bayes	0.64	0.83	0.83	51.56	94.92
Support Vector Machine	0.98	0.93	**0.93**	52.21	96.37
Logistic Regression	0.99	**0.94**	0.93	**49.96**	94.70
Random Forest	**0.99**	0.91	0.90	63.11	96.89
Maxent	0.99	0.94	0.92	1872.08	**20.34**

Looking at Table 5.3, in the training phase, only Naive Bayes fails to learn and fit the HT class, with a precision of 31%. Conversely, all other algorithms, except Naive Bayes, achieve good precision results in the testing phase, generally exceeding 90%. High precision values indicate that they correctly identify positive cases. However, this is not the whole story.

Table 5.3: Summary of precision from ML models with emoji features. *Note: the average column is for the macro average.

Model	Train			Test 1			Test 2		
	Sexual	HT	avg	Sexual	HT	avg	Sexual	HT	avg
Naive Bayes	1.00	0.31	0.65	0.96	0.55	0.76	0.96	0.50	0.73
Support Vector Machine	0.97	0.99	0.98	0.93	0.97	**0.95**	0.92	0.93	**0.93**
Logistic Regression	0.99	1.00	**0.99**	0.94	0.96	**0.95**	0.93	0.93	**0.93**
Random Forest	1.00	0.99	**0.99**	0.92	0.94	0.93	0.90	0.92	0.91
Maxent	0.99	0.99	**0.99**	0.93	0.95	0.94	0.92	0.92	**0.93**

Upon reviewing Table 5.4, it becomes evident that most algorithms experience a drop in recall performance in the testing phase for the HT class. This indicates that they are missing many positive cases related to human trafficking (HT). This issue may arise from the imbalanced nature of the data, with these machine learning models leaning more toward the Sexual class and less towards the HT class. Given that the problem's nature reflects the data distribution in the real world, where there are few tweets related to human trafficking, this study must identify which model can adapt to this data imbalance effectively.

Table 5.4: Recall metric from ML models with emoji features. *Note: the average column is for the macro average.

Model	Train			Test 1			Test 2		
	Sexual	HT	avg	Sexual	HT	avg	Sexual	HT	avg
Naive Bayes	0.60	0.99	0.79	0.88	0.82	**0.85**	0.86	0.79	**0.83**
Support Vector Machine	1.00	0.85	0.92	1.00	0.58	0.79	0.99	0.50	0.75
Logistic Regression	1.00	0.95	0.97	1.00	0.67	0.83	0.99	0.56	0.78
Random Forest	1.00	0.99	**0.99**	0.99	0.52	0.76	0.99	0.41	0.70
Maxent	1.00	0.93	0.97	1.00	0.58	0.79	0.99	0.49	0.74

Additionally, upon reviewing Table 5.5, it is clear that most ML models need help to classify the HT class accurately. The test sets show subpar performance overall. The ML model needs to be better equipped to handle this limitation due to the imbalanced nature of the original dataset. The models are unable to balance all the metrics effectively, and without analyzing all metrics, it may appear as though the models have excellent performance. This is why we have chosen to analyze all metrics.

Table 5.5: F1-score metric from ML models with emoji features. *Note: the average column is for the macro average.

Model	Train			Test 1			Test 2		
	Sexual	HT	avg	Sexual	HT	avg	Sexual	HT	avg
Naive Bayes	0.75	0.47	0.61	0.92	0.66	0.79	0.91	0.61	0.76
Support Vector Machine	0.99	0.92	0.95	0.96	0.72	0.84	0.95	0.65	0.80
Logistic Regression	0.99	0.97	0.98	0.97	0.79	**0.88**	0.96	0.70	**0.83**
Random Forest	1.00	0.99	**0.99**	0.96	0.67	0.81	0.95	0.57	0.76
Maxent	0.99	0.96	0.98	0.96	0.72	0.84	0.95	0.64	0.80

5.3 LSTM-based algorithms

Table 5.6: Summary of training 5-fold cross-validation of LSTM-based models

Model	Train				Time (s)
	F1-score	Precision	Recall	Accuracy	Train
LSTM (w2v)	0.82	0.89	0.76	0.95	232.12
LSTM (w2v+e2v)	0.84	0.91	0.78	**0.96**	245.80
Bi-LSTM (w2v)	0.83	0.89	0.78	0.95	356.51
Bi-LSTM (w2v+e2v)	0.84	0.92	0.78	**0.96**	370.22
C-LSTM (w2v)	0.84	0.90	0.80	0.95	171.15
C-LSTM (w2v+e2v)	0.79	0.79	0.79	0.93	199.96
LSTM (ada-002)	0.76	0.76	0.76	0.93	164.77
LSTM (ada-002 + DA)	0.93	0.93	**0.93**	0.94	123.63
Bi-LSTM (ada-002)	0.78	0.80	0.76	0.94	287.67
Bi-LSTM (ada-002 + DA)	**0.94**	**0.95**	0.93	0.95	159.17
C-LSTM (ada-002)	0.74	0.92	0.62	0.93	105.24
C-LSTM (ada-002 + DA)	0.92	0.93	0.91	0.93	**92.93**

Table 5.7: Summary of testing 5-fold cross-validation of LSTM-based models

Model	Test				Time (s)
	F1-score	Precision	Recall	Accuracy	Eval
LSTM (w2v)	0.71	0.77	0.67	0.92	0.85
LSTM (w2v+e2v)	0.73	0.80	0.68	**0.93**	1.04
Bi-LSTM (w2v)	0.73	0.78	0.69	0.92	1.42
Bi-LSTM (w2v+e2v)	0.72	0.79	0.67	0.92	2.35
C-LSTM (w2v)	0.72	0.76	0.69	0.92	0.60
C-LSTM (w2v+e2v)	0.71	0.70	0.73	0.91	0.79
LSTM (ada-002)	0.72	0.73	0.72	0.92	0.63
LSTM (ada-002 + DA)	0.91	0.91	**0.91**	0.92	0.74
Bi-LSTM (ada-002)	0.72	0.74	0.71	0.92	1.19
Bi-LSTM (ada-002 + DA)	**0.91**	**0.92**	0.89	**0.93**	1.31
C-LSTM (ada-002)	0.70	0.87	0.59	**0.93**	**0.41**
C-LSTM (ada-002 + DA)	0.90	0.91	0.89	0.92	0.44

The experiments conducted in this study involved LSTM-based algorithms, which were divided into three categories. The first category used Word2Vec (w2v) as embeddings, the second category used Word2Vec plus Emoji2Vec (w2v + e2v), and the third category used ada-002 embeddings (ada-002), which were further divided into models that included data augmentation (DA) and those that did not (see Table 5.7).

The results in Table 5.7 showed that the Bi-LSTM models outperformed the other two architectures regarding f1-score, precision, recall, and accuracy. For example, the Bi-LSTM (w2v) model had an f1-score of 73.3%, precision of 78.1%, recall of 69.2%, and accuracy of 92.4%. It outperformed its direct counterparts, such as LSTM (w2v) and C-LSTM (w2v), by a few points. Similarly, the Bi-LSTM (ada-002) model outperformed its counterparts, which are LSTM (ada-002) and C-LSTM (ada-002) models, in terms of f1-score, precision, recall, and accuracy.

Models that included emojis, such as all those that used ada-002 and w2v+e2v, obtained better and more consistent results during the training and testing phases. This may be due to their ability to capture more relevant information, which helped them identify patterns related to the label and the tweets in question.

Furthermore, models that used data augmentation, such as LSTM (ada-002 + DA), Bi-LSTM (ada-002 + DA), and C-LSTM (ada-002 + DA), significantly outperformed the models that did not, given that the dataset was highly imbalanced. This shows that this limited the other models without data augmentation and that the use of Large Language Models as data generation engines allows us to overcome this data limitation.

The Bi-LSTM (ada-002 + DA) model had the highest performance in training in terms of f1-score, precision, and accuracy, with 84.4%, 92.0%, and 95.6%, respectively. This model could avoid false positives and had a good balance between precision and recall,

indicating that it is a balanced model in terms of performance.

On the other hand, the C-LSTM (w2v+e2v) model, in the testing section, outperformed other models in the recall metric with 72.7%. This indicates that it predicted more true positives (HT class) than others. Still, it was the model with the lowest performance in precision at 69.9%, suggesting that it was the worst at avoiding false positives.

The Bi-LSTM (no emoji) model had a higher f1-score, indicating a balance between recall and precision. Despite not including emojis, it had an f1-score of 73.3%, suggesting it maintained a balance between correctly identifying positive cases (precision) and its ability to recognize the most favorable cases.

Additionally, the LSTM 2-stream achieved 95.6% accuracy in the training phase and 92.5% in the testing phase, making it the model with the highest accuracy. However, upon reviewing other metrics, we observe that the model has the highest precision compared to others at 79.7%

The C-LSTM (ada-002 + DA) models were the fastest to train, taking 92.93 seconds despite using data augmentation, which logically would mean they take longer due to more input data. This could be due to the lightness of these models, specifically, that this model uses very light and optimized embeddings such as those from OpenAI. The Bi-LSTM models took the longest to train and perform inference.

Finally, the results indicate that the improved performance of the Bi-LSTM could be due to its bidirectional capability to relate words, which is why it takes longer in training and inference. On the other hand, the C-LSTM models are fast for training and inference, which would allow us to use them in cases where real-time inference is required. Finally, the LSTM models are the most balanced in performance training and inference time. Therefore, these models serve as a comparative base for others and could be used for more studies on this subject. They also show that often, the power is not only in the model itself but in the combination of factors that affect the model's performance.

Chapter 6

Conclusions

The conclusions drawn from this research work indicate that LSTM-based models perform very well in classification cases when they are correctly configured and when good embedding techniques are employed. In this specific case, including emojis as a feature enhances the overall model performance. Machine learning models were impractical for this problem, mainly due to the imbalanced dataset. They need help to learn the class with fewer samples and make numerous errors in identifying true positives, leading to low recall metrics.

It is worth mentioning that LSTM-based models take significantly longer to train than machine learning models, approximately ten times more. However, they are much faster in batch processing during inference than machine learning models. Furthermore, LSTM-based models outperform machine learning models significantly. Several factors contribute to this, including text representation, preprocessing, and the robustness of LSTM models, specifically designed for text processing and excel at capturing word relationships through embeddings.

Although traditional ML models like Naive Bayes, SVM, and Logistic Regression differ fundamentally from LSTM in their architecture, they are comparable in terms of classification performance on the same dataset. The classical models rely on static, non-sequential features, whereas LSTM captures contextual and sequential patterns. This comparison serves to benchmark the effectiveness of deep learning against conventional approaches within the same classification task.

A surprising finding is that convolutional models are much faster for training and inference, demonstrating that combining deep learning techniques can lead to faster and more efficient models without sacrificing performance. On the other hand, Bi-LSTM models take longer to train and infer but yield better overall performance. This provides a range of possibilities for model selection depending on the specific use case.

Another important aspect was the creation of the dataset with the corresponding labels. This shows us that, for now, human intervention is necessary to create this type of dataset, given that little information and research allows us to label accurately and safely in an automated manner.

Another conclusion for this work is that we tested different types of text representation, such as Bag of Word, Word2Vec, Emoji2Vec, and ada-002. This allowed us to experiment with the fact that often, only a good representation of the text is necessary for the model to

perform better. This was evident when the models using Word2Vec + Emoji2Vec achieved better performance than those that only considered the text or the number of words. Moreover, new models for embeddings like ada-002 manage to vectorize the entire text set into a single vector, capturing these semantics since they are used for Large Language Models.

Additionally, an important aspect was to observe that LLMs can help us solve the problem of imbalance by generating new tweets with similar words based on those we already had. This allowed us to overcome the limit that the models had to learn due to the significant class imbalance of the data, achieving a considerable improvement.

Finally, an important conclusion is that we collected relevant tweets that allowed us to build a dataset that works despite an imbalance but shows us a reality. This problem will continue to exist despite efforts to combat it. The implementation of artificial intelligence techniques, particularly deep learning models such as LSTM, enables the automated analysis of social media content to detect subtle indicators of human trafficking. This study contributes by providing a context-aware approach specifically designed for the Spanish-speaking digital landscape, where such tools are particularly scarce. The model developed can be integrated into monitoring systems used by Non-Governmental Organizations, law enforcement agencies, or social media platforms to automatically identify content indicative of illicit activities. By detecting linguistic patterns and behavioral indicators associated with trafficking-related activities, the system supports early intervention and resource allocation, ultimately enhancing preventive strategies in the fight against human trafficking.

For future research, the dataset could be expanded with additional annotated tweets to increase representativeness and robustness. Retraining the models with a larger and more diverse dataset may enhance their generalization capabilities and performance. Furthermore, the issue of class imbalance (an inherent challenge in human trafficking detection) could be addressed more systematically through advanced resampling strategies or cost-sensitive learning, where greater weight is assigned to the minority class during training.

Another promising direction involves reframing the task as an anomaly detection problem, due to the rarity and subtlety of trafficking-related content. By training models primarily on common, lawful sexual content, it may be possible to identify outliers that exhibit unusual or suspicious linguistic and behavioral patterns. This alternative formulation could open the door to unsupervised or semi-supervised approaches, offering novel solutions for scenarios where labeled data is limited or sensitive. Exploring this direction could significantly enrich the toolkit available for social media-based trafficking detection.

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